Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/13/2012
PI Name:	Mollicone, Daniel Ph.D.		
Project Title:	Individualized Behavioral Health Monitoring Tool		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	06/01/2011	**End Date:**	03/31/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Given the extended duration of future missions and the isolated, extreme and confined environments, there is the possibility that behavioral conditions and mental disorders (DSM-IV-TR) will develop. The overarching goal of this project is to deliver an Individualized Behavioral Health Monitoring Tool that unobtrusively integrates all available behavioral measures collected during a mission to provide a dashboard of behavioral health indicators. These indicators will be placed within the context of quantitatively-tracked mission stressors to provide meaningful feedback allowing behavioral issues to be detected and mitigated at an early stage. The result of this project will be a system prototype that can be deployed in space analog environments (e.g., Mars 520 study, Antarctica) for validation testing and ultimately deployed in long-duration space exploration missions. The critical need for an Individualized Behavioral Health Monitoring Tool has been identified as a priority outlined in the BHP Integrated Research Plan (July 2009) gap BMED3. Phase II will achieve: (1) an Individualized Behavioral Health Monitor software interface; (2) a data		

Task Description:	integration system; (3) a trend and change detection algorithm; and (4) a countermeasure selection aid. The Technology Readiness Level at the end of Phase II will be TRL 6. POTENTIAL NASA COMMERCIAL APPLICATIONS: The Individualized Behavioral Health Monitoring Tool will meet the specific requirements of long duration exploration missions and provide feedback to astronauts, Op Psy Personnel and Flight Surgeons about behavioral health status as well as aid in the selection of countermeasures. It will be designed to be unobtrusive and require minimal crew time or effort to train and utilize. The resulting product will be primarily relevant to NASA's Behavioral Health and Performance (BHP) research gaps (as of July 2009): BMED 3 (What are the optimal methods to detect and assess decrements in behavioral health during exploration missions?) but will also be relevant to gaps BMED1, BMED2, BMED6, BMED7, and BMED8. When validated, the Individualized Behavioral Health Monitoring Tool will be deployed in long-duration space exploration missions to support crew behavioral health efforts during training, mission and return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Individualized Behavioral Health Monitoring Tool can be adapted to meet an articulated need to track behavioral health in occupations associated with high stress, high workload, and high danger factor such as military applications and law enforcement. A tool that enables the systematic and efficient tracking of individual behavioral health status in these occupational settings can provide a means to detect and address behavioral disorders and mental conditions at an early stage. Taking military operations as an example, there is evidence that behavioral disorders and mental conditions such as depression, post-traumatic stress disorder, and traumatic brain injury have a high prevalence among soldiers. There is a present market opportunity to deliver an Individualized Behavioral Health Monitoring Tool to track changes in behavioral health status in soldiers during training, deployment, and post-deployment. The Army currently has 238,000 soldiers deployed overseas in 120 countries (source: US Army, 2006).
Task Progress:	New project for FY2011. Reporting not required for this SBIR Phase 2 project. (Ed. note: added to Task Book 9/13/2012 when received information)
Bibliography Type:	Description: (Last Updated: 02/23/2015)